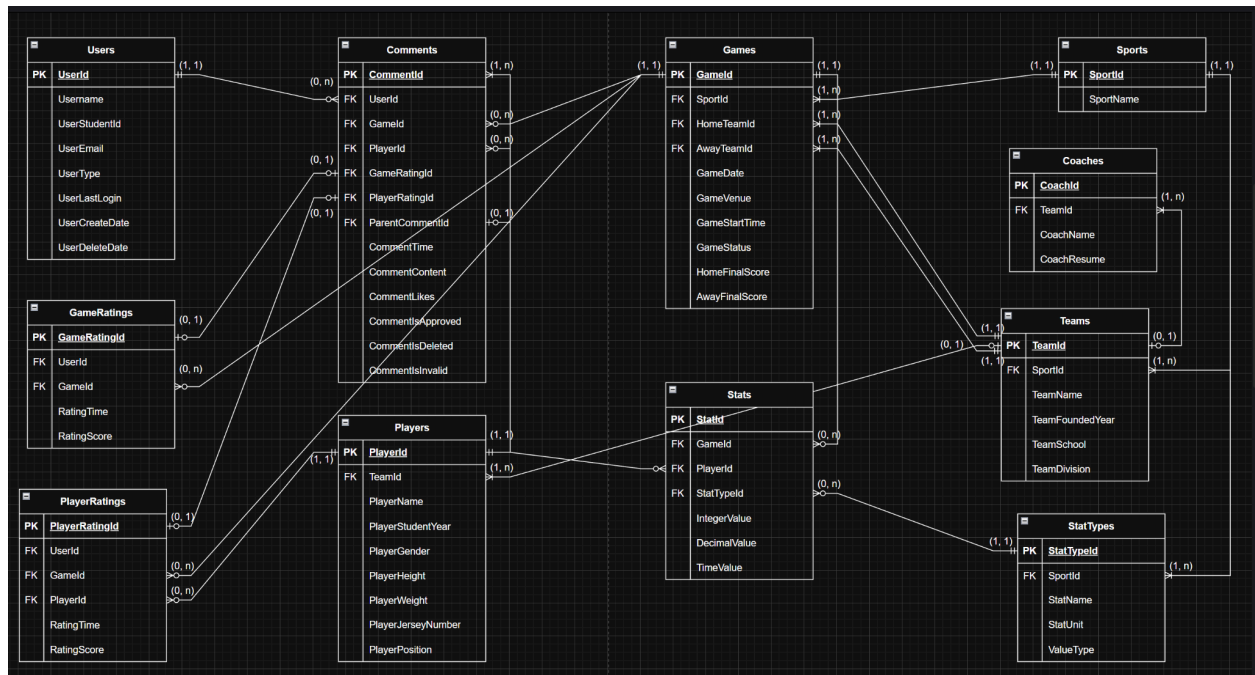


LIS 464: Applied Database Design - NoSQL Project

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Part 0: Original SQL Design

Original SQL ERD:



Part 1: Design

Design 1: Array Implementation (Teams & Coaches)

3.1 Screenshots:

lis464.reywwyg.mongodb.net > Final_Project > Teams Open MongoDB shell

Documents 2 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 2 of 2 ↺ ↻ ⌵ ⌶ ⌷

```

_id: 101
TeamName: "Wisconsin Badgers"
TeamFoundedYear: 1848
TeamSchool: "University of Wisconsin-Madison"
TeamDivision: "NCAA Division I"
SportId: 1
CoachNames: Array (4)
  0: "Greg Gard"
  1: "Joe Krabbenhoft"
  2: "Dean Oliver"
  3: "Sharif Chambliss"
Players: Array (2)

```

```

_id: 102
TeamName: "Purdue Boilermakers"
TeamFoundedYear: 1869
TeamSchool: "Purdue University"
TeamDivision: "NCAA Division I"
SportId: 1
CoachNames: Array (4)
  0: "Matt Painter"
  1: "Brandon Brantley"
  2: "Terry Johnson"
  3: "Paul Lusk"
Players: Array (2)

```

3.2 Explanation for Design 1 "array":

- Relationship Analysis:** This design models a **1:M relationship** between a **Team (1 side)** and **Coaches (M side)**. Instead of creating a separate "Coaches" collection and joining them, we embed the list of coach names directly into the Team document.
- Array Limit:** The values in the CoachNames array are strictly limited by real-world constraints. A basketball team has a finite number of coaching staff members allowed by regulations (typically 1 Head Coach and 3-4 Assistants)⁷.
- What is the limit?** The limit is the NCAA regulation cap on coaching staff, ensuring this array will never become "unbounded" or infinite.

3.3 JSON Document (Sample):

(See attached file design1_team_array.json)

```

{
  "_id": 101,
  "TeamName": "Wisconsin Badgers",
  "TeamFoundedYear": 1848,
  "TeamSchool": "University of Wisconsin-Madison",
  "TeamDivision": "NCAA Division I",

```

```

    "SportId": 1,
    "CoachNames": [
      "Greg Gard",
      "Joe Krabbenhoft",
      "Dean Oliver",
      "Sharif Chambliss"
    ],
    "Players": [1001, 1002]
  }

```

Design 2: Subdocuments Implementation (Sports & Rules)

4.1 Screenshots:

The screenshot shows the MongoDB Atlas web interface for a database named 'Final_Project' in the 'Sports' collection. The interface displays two documents:

- Document 1 (Basketball):**

```

{
  "_id": 1,
  "SportName": "Basketball",
  "LeagueStructure": {
    "GoverningBody": "NCAA",
    "PrimaryDivision": "Division I",
    "SeasonFormat": "Winter",
    "ChampionshipType": "March Madness"
  },
  "EquipmentRules": {
    "BallSize": "29.5 inches",
    "RimHeight": "10 feet"
  }
}

```
- Document 2 (Football):**

```

{
  "_id": 2,
  "SportName": "Football",
  "LeagueStructure": {
    "GoverningBody": "NCAA",
    "PrimaryDivision": "FBS (Football Bowl Subdivision)",
    "SeasonFormat": "Fall",
    "ChampionshipType": "CFP Playoffs"
  },
  "EquipmentRules": {
    "BallSize": "Official Collegiate Size",
    "FieldLength": "100 yards"
  }
}

```

4.2 Explanation for Design 2 "subdocuments":

- Relationship Analysis:** This represents a relationship where **Sport (1 side)** contains specific details about its **League Structure (M side/Child Entity)**.
- By embedding LeagueStructure and EquipmentRules as subdocuments, we group strict 1:1 details (specific to that sport) directly with the parent object for faster read performance.

4.3 JSON Document (Sample):

(See attached file design2_sport_subdoc.json)

```
{
  "_id": 1,
  "SportName": "Basketball",
  "LeagueStructure": {
    "GoverningBody": "NCAA",
    "PrimaryDivision": "Division I",
    "SeasonFormat": "Winter",
    "ChampionshipType": "March Madness"
  },
  "EquipmentRules": {
    "BallSize": "29.5 inches",
    "RimHeight": "10 feet"
  }
}
```

Design 3: Subdocuments in an Array (Players & Stats)

5.1 Screenshots:

lis464.reywwyg.mongodb.net > Final_Project > Players Open MongoDB shell

Documents 4 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 4 of 4 ↺ ↻ ⌵ ☰ { } ⌂

```
_id: 1001
teamId: 101
PlayerName: "Chuckie Hepburn"
PlayerJerseyNumber: 23
PlayerPosition: "Guard"
PlayerHeight: "6'2"
PlayerWeight: 205
PlayerStudentYear: "Junior"
PlayerGender: "M"
stats: Array (2)
```

```
_id: 1002
teamId: 101
PlayerName: "Steven Crawl"
PlayerJerseyNumber: 22
PlayerPosition: "Forward"
PlayerHeight: "7'0"
PlayerWeight: 247
PlayerStudentYear: "Junior"
PlayerGender: "M"
stats: Array (2)
  0: Object
    Season: "2022-2023"
    PointsPerGame: 12.1
    AssistsPerGame: 2.5
    ReboundsPerGame: 6.9
  1: Object
    Season: "2023-2024"
    PointsPerGame: 11.2
    AssistsPerGame: 2.1
    ReboundsPerGame: 7.3
```

5.2 Explanation for Design 3 "subdocuments in an array":

- **Relationship Analysis:** This models a **1:M relationship** between a **Player (1 side)** and their **Seasonal Statistics (M side)**.
- A player participates in multiple seasons. Each season has a distinct set of metrics (Points, Assists, Rebounds). We embed these as objects inside the stats array because they belong exclusively to that player and are frequently accessed together with the player's profile.

5.3 JSON Document (Sample):

(See attached file design3_players_subdocarray.json)

```
{
  "_id": 1001,
  "teamId": 101,
  "PlayerName": "Chucky Hepburn",
  "PlayerJerseyNumber": 23,
  "PlayerPosition": "Guard",
  "stats": [
    {
      "Season": "2022-2023",
      "PointsPerGame": 12.2,
      "AssistsPerGame": 2.8,
      "ReboundsPerGame": 2.8
    },
    {
      "Season": "2023-2024",
      "PointsPerGame": 11.5,
      "AssistsPerGame": 3.9,
      "ReboundsPerGame": 3.3
    }
  ]
}
```

Part 2: Schemas and Queries

JSON Schema (Design 3: Players)

Schema Title: Player Performance Schema

Description: Validation rules for player documents containing personal details and nested statistical records.

Validation Elements Included:

- **Required Elements:** PlayerName, PlayerJerseyNumber, PlayerPosition, teamId, stats.
- **Limitations/Rules:**
 - PlayerJerseyNumber: Must be an integer between 0 and 99.
 - PlayerPosition: Must be an enum of "Guard", "Forward", or "Center".
 - PointsPerGame: Must be a minimum of 0.

Validation Proof:

lis464.reywwyg.mongodb.net > Final_Project > Players

Open MongoDB shell

Documents 4 Aggregations Schema Indexes 1 Validation

67 properties: {
68 Season: {
69 bsonType: 'string',
70 description: 'The specific academic year of the stats (e.g., 2023-2024).'
71 },
72 PointsPerGame: {
73 bsonType: 'double',
74 minimum: 0,
75 description: 'Average points scored per game. Cannot be negative.'
76 },
77 AssistsPerGame: {
78 bsonType: 'double',
79 minimum: 0,
80 description: 'Average assists per game. Cannot be negative.'
81 },
82 ReboundsPerGame: {
83 bsonType: 'double',
84 minimum: 0,
85 description: 'Average rebounds per game. Cannot be negative.'
86 },
87 },
88 },
89 },
90 },
91 },
92 }

Edit rules

Passed validation Failed validation

```
_id: 1002  
teamId: 101  
PlayerName: "Steven Crowl"  
PlayerJerseyNumber: 22  
PlayerPosition: "Forward"  
PlayerHeight: "7'0"  
PlayerWeight: 247  
PlayerStudentYear: "Junior"  
PlayerGender: "M"  
stats: Array (2)
```

No Preview Documents

Queries

1. Greater than or less than query:

Purpose: Find all players weighing more than 200 lbs.

```
db.players.find({ "PlayerWeight": { $gt: 200 } })
```

lis464.reywwyg.mongodb.net > Final_Project > Players Open MongoDB shell

Documents 4 Aggregations Schema Indexes 1 Validation

`{ "PlayerWeight": { $gt: 200 } }` Generate query Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 2 of 2 < > ≡ {} ⌂

```
{
  "_id": 1001,
  "teamId": 101,
  "PlayerName": "Chucky Hepburn",
  "PlayerJerseyNumber": 23,
  "PlayerPosition": "Guard",
  "PlayerHeight": "6'2\"",
  "PlayerWeight": 205,
  "PlayerStudentYear": "Junior",
  "PlayerGender": "M",
  "stats": Array (2)
}
```

```
{
  "_id": 1002,
  "teamId": 101,
  "PlayerName": "Steven Crowl",
  "PlayerJerseyNumber": 22,
  "PlayerPosition": "Forward",
  "PlayerHeight": "7'0\"",
  "PlayerWeight": 247,
  "PlayerStudentYear": "Junior",
  "PlayerGender": "M",
  "stats": Array (2)
}
```

2. Query within an array:

Purpose: Find the team that has "Greg Gard" listed in their coaches array.

```
db.teams.find({ "CoachNames": "Greg Gard" })
```

lis464.reywwyg.mongodb.net > Final_Project > Teams Open MongoDB shell

Documents 2 Aggregations Schema Indexes 1 Validation

`{ "CoachNames": "Greg Gard" }` Generate query Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 1 of 1 < > ≡ {} ⌂

```
{
  "_id": 101,
  "TeamName": "Wisconsin Badgers",
  "TeamFoundedYear": 1848,
  "TeamSchool": "University of Wisconsin-Madison",
  "TeamDivision": "NCAA Division I",
  "SportId": 1,
  "CoachNames": Array (4)
    0: "Greg Gard"
    1: "Joe Krabbenhoft"
    2: "Dean Oliver"
    3: "Sharif Chambliss"
  "Players": Array (2)
}
```

3. Query within a subdocument:

Purpose: Find sports where the season format (inside the LeagueStructure subdocument) is "Winter".

```
db.sports.find({ "LeagueStructure.SeasonFormat": "Winter" })
```

Documents 2 Aggregations Schema Indexes 1 Validation

{ "LeagueStructure.SeasonFormat": "Winter" }

Generate query

Explain

Reset

Find

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

25

1 - 1 of 1

```
{
  "_id": 1,
  "SportName": "Basketball",
  "LeagueStructure": {
    "GoverningBody": "NCAA",
    "PrimaryDivision": "Division I",
    "SeasonFormat": "Winter",
    "ChampionshipType": "March Madness"
  },
  "EquipmentRules": {
    "BallSize": "29.5 inches",
    "RimHeight": "10 feet"
  }
}
```